1. Objectives:

- To understand the concept of the Fixed-point iteration Method using python.
- To learn how to approximate the root of the nonlinear equation.

2. Introduction:

The Iteration method or the Fixed-point iteration Method is a numerical technique for solving nonlinear equations of the form f(x) = 0. the equation is rearranged into the form: x = g(x) and starting from an initial guess x0, the process continues until the difference becomes smaller than a given tolerance. the convergence depends on the choice of g(x).

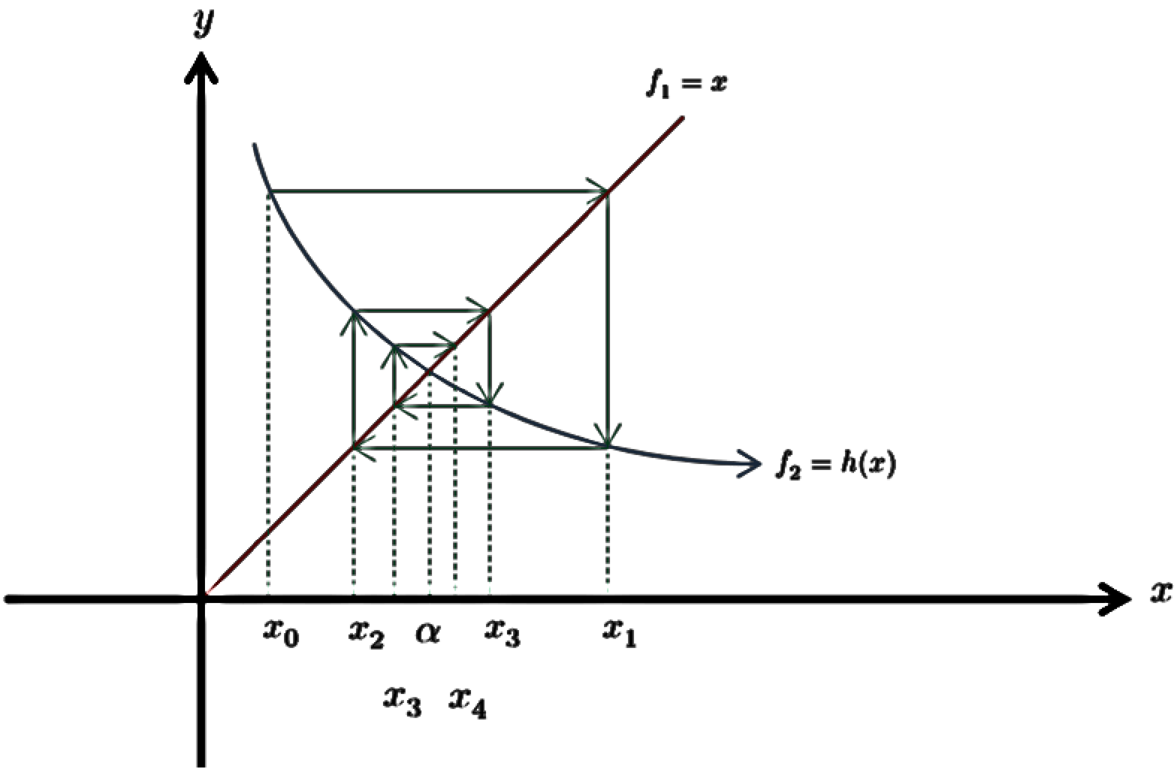

Fig 1: Fixed-point iteration Method

3. Procedure

Algorithm: Fixed-point iteration Method in Python

1. Start with an initial x0.
2. Compute x1 = g(x0).
3. Check if it is less than tolerance, if yes then stop.
4. Otherwise, set x0 = x1 and repeat.
5. Continue until the root is found within tolerance or maximum iterations are reached.

4. Required Hardware and Software:

- **Hardware:** AMD RYZEN 7, RAM 8 GB
- **Software:** PyCharm Community Edition 20251.1.1

5. Experiment Implementation:

Code:

```
import math

def f(x):

    return 3 * x - math.cos(x) - 1

def g(x):

    return (1 + math.cos(x)) / 3

x0 = 3.2

tol = 0.0005


def iteration(x0, tol, max_iter=100):

    print(f"{'Iteration':<12}{'x':<12}{'f(x)':<12}")

    print("-" * 40)

    i = 1

    while i <= max_iter:

        x1 = g(x0)

        fx = f(x1)

        print(f"{i:<12}{x1:<12.6f}{fx:<12.6f}")


        if abs(x1 - x0) < tol:

            print(f"\nRoot is approximately: {x1:.6f}")

            return

        x0 = x1

        i += 1

    print("\nDid not converge within max iterations.")

iteration(x0, tol)
```

Output:

```
Iteration        x          f(x)
----------------------------------------
1          0.000568    -1.998295
2          0.666667    0.214113
3          0.595296    -0.042095
4          0.609328    0.007950
5          0.606678    -0.001514
6          0.607182    0.000288
7          0.607086    -0.000055

Root is approximately: 0.607086
```

5. Discussion & Conclusion:

The Fixed-Point Iteration Method successfully approximated the root of the given nonlinear equation. Starting with an initial guess of x0 = 3.2 the method quickly converged to the root near x ~ 0.604 within the specified tolerance. The iteration table shows how each step brought the approximation closer to the solution.